

HIRUNA VISHWAMITH

[github/hiruna](#) ◇ [linkedin.com/hiruna](#) ◇ [portfolio/hiruna](#) ◇ [scholar/hiruna](#) ◇ hvishwamith@gmail.com

EDUCATION

University of Moratuwa, Sri Lanka

Jun 2021 - Jul 2025

- BSc. (Hons) in Electronics and Telecommunications Engineering
- Minor in Mathematics
- GPA: 3.75/4.0, Deans List 1st, 2nd, 6th, 8th Semesters

FINAL YEAR PROJECT

Cache coherent, Out of Order, Multi-core RISC-V CPU

- Developed a parametric multi-core RISC-V emulator capable of running bare-metal multi-core programs and booting Linux in multi-core mode.
- Developed an Instruction Fetch Unit (IFU) with a TAGE branch predictor, which is capable of achieving the desired performance goals on an FPGA.
- Developed a multi-threading benchmark suite for benchmarking and testing multi-core RISC-V CPU
- Implemented a Cache Coherent, Out of Order, Multi-core RISC-V CPU on FPGA capable of booting Linux in multi-core mode.
- Developed an end-to-end, cache-coherent, out-of-order, multi-core RISC-V processor development platform, capable of cycle-accurate cross-validation between a multicore emulator and simulator, thereby ensuring faster debugging times.
- Currently **recognized as Sri Lanka's first academic multicore processor and debugging platform.**

PUBLICATIONS

- T. Coskun, **H. Vishwamith**, M. Isik and I. Can Dikmen, "Hardware-Accelerated Transformer Framework for Real-Time Battery SoH Estimation," IEEE High Performance Extreme Computing Conference (HPEC), 2025. [Link to :[Paper](#)]
- T. Coskun, **H. Vishwamith**, M. Isik and I. Can Dikmen, "Exploring Neuromorphic Computing with Loihi-2 for High-Performance CFD Simulations," IEEE High Performance Extreme Computing Conference (HPEC), 2025. [Link to :[Paper](#)]
- T. Coskun, **H. Vishwamith**, M. Isik and I. Can Dikmen, "Symbolic Computing with Spiking Neural Networks on GSM8K," IEEE International Conference on Microelectronics (ICM), 2025. (Under review)
- **H. Vishwamith**, T. Coskun, M. Isik and I. Can Dikmen, "A Reconfigurable Framework for AI-FPGA Agent Integration and Acceleration," IEEE International Conference on Microelectronics (ICM), 2025. (Under review)
- **H. Vishwamith**, L. Fernando, M. Rathnayake, Y. Amarashinghe, K. Yalgama and A. Pasquel, "An End to End Multicore Processor Development Platform for Teaching and Learning Computer Architecture in Higher Education," IEEE Transactions on Education (TOE), 2025. (Under review)
- **H. Vishwamith**, M. Isik and I. Can Dikmen, "HPCNeuroNet: Advancing Neuromorphic Audio Signal Processing with Transformer-Enhanced Spiking Neural Networks," IEEE Interdisciplinary Conference on Electronics and Computer (INTCEC), 2024. [Link to : [Paper](#)]
- M. Isik, **H. Vishwamith**, Y. Sur and I. Can Dikmen, "NEUROSEC: FPGA-Based Neuromorphic Audio Security," International Symposium on Applied Reconfigurable Computing (ISARC), 2024. [Link to :[Paper](#)]
- M. Isik, **H. Vishwamith**, J. Naoukin and I. Can Dikmen, "HPCNeuroNet: A Neuromorphic Approach Merging SNN Temporal Dynamics with Transformer Attention for FPGA-based Particle Physics," Conference on Computing in High Energy and Nuclear physics (CHEP), 2024. [Link to : [Paper](#)]

EXPERIENCE

FPGA Design Engineer

Apex Compute

Mar 2025 - present

USA, California

- ❖ Designing and implementing energy-efficient, domain-specific architectures on FPGA for LLMs.
- ❖ Achieve 2x throughput on FPGA than, NVIDIA Jetson Orin.

Research Scholar - Internship

National University of Singapore

Dec 2023 - May 2024

Singapore, Clementi

- ❖ Designed and implemented 3DRA CGRA (Coarse-Grained Reconfigurable Architecture) on U250 FPGA, achieving a 650MHz operating frequency.
- ❖ Optimized data communication for a 100Gbps TCP/IP Offload Engine.

Project Assistant

University of Moratuwa

Jan 2023 - Apr 2023

Sri Lanka, Moratuwa

- ❖ Contributed to developing an Intelligent Camera with Real-Time Video Analytics System.
- ❖ Developed a License Plate Recognizing web app using YOLOv5.

[\[web app\]](#)

TECHNICAL SKILLS

Hardware & Computer Architecture:

Verilog, RISC-V, FPGA, Cache Coherency, Parallel Computing

Development Tools:

Xilinx Vivado, Intel Quartus, Vitis, Verilator

Programming:

Python, C++, C, Chisel

Machine Learning & AI Acceleration:

PyTorch, TensorFlow, CUDA

COMPETITIONS

ACM/IEEE TinyML Design Contest @ ICCAD 2023

Sep 2023 - Nov 2023

- ❖ Implementing deep learning algorithm that can automatically discriminate VAs (i.e., Ventricular Fibrillation and Ventricular Tachycardia) [\[leaderboard\]](#)
- ❖ Deployed model on STM32F303K8T6 (ARM Cortex M4).
- ❖ Optimized inference latency to achieve 2nd **place in latency score** with latency 0.52ms

IEEE Signal Processing Society VIP Cup 2023

Aug 2023 - Sep 2023

- ❖ Developed a ML model using SIMCLR for VIP Cup 23 - IEEE Signal Processing Society [\[leaderboard\]](#)
- ❖ The ML model was capable of identifying 6 Biomarkers of OCT images using clinical data
- ❖ I was able to achieve F1 score of 70% in the final phase